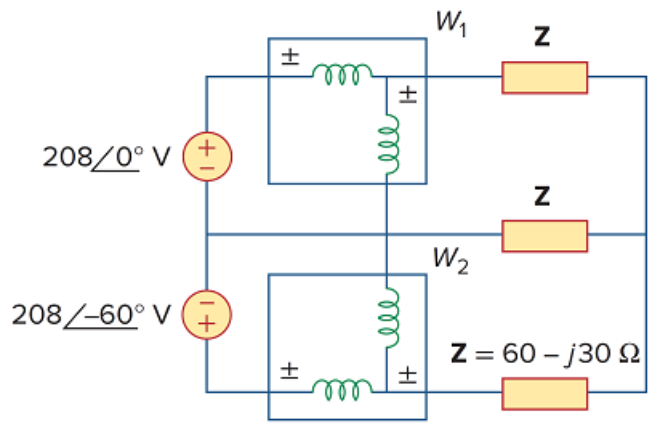


Problem

Predict the wattmeter readings for the circuit in Fig. 12.75.

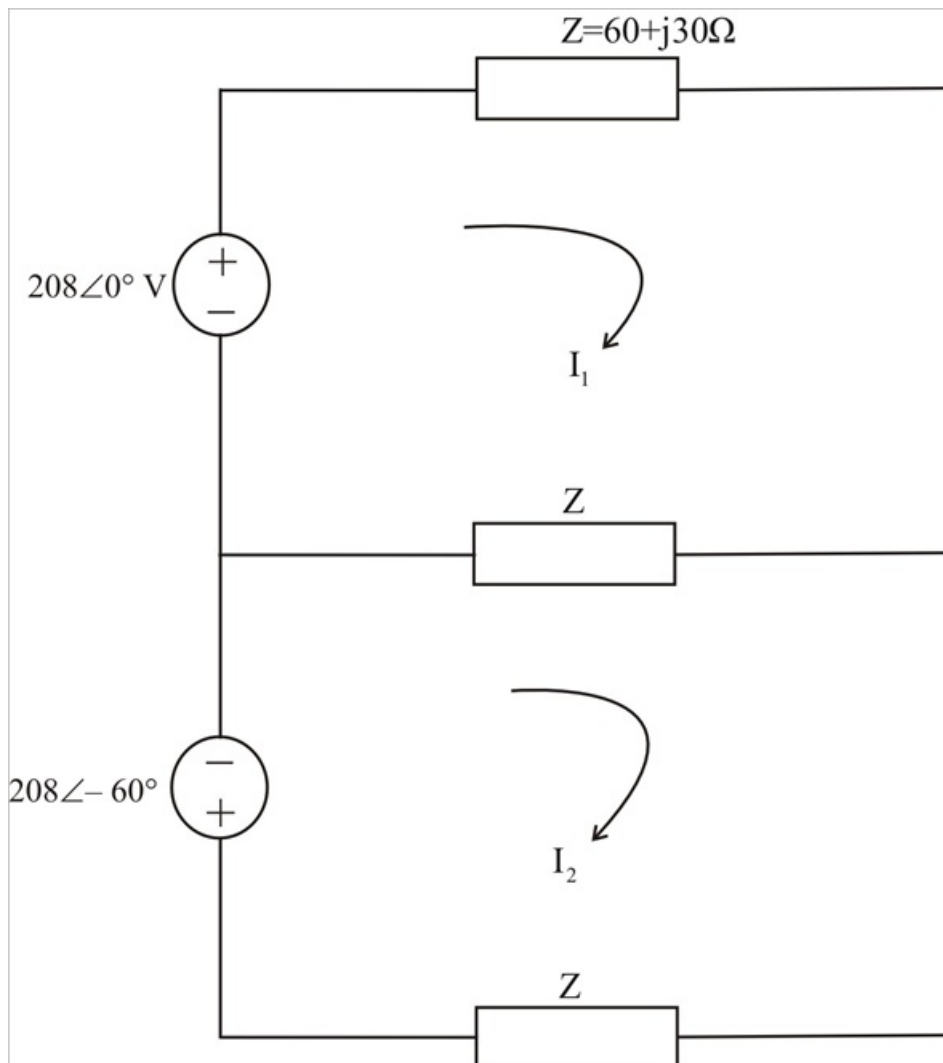
Figure 12.75



Step-by-step solution

Step 1 of 2

Consider the circuit below.



Step 2 of 2

For Mesh 1,

$$208 = 2Z_1 - ZI_2$$

For Mesh2,

$$-208\angle -60^\circ = -Z_1I_1 + 2ZI_2$$

In matrix form,

$$\begin{bmatrix} 208 \\ 208\angle -60^\circ \end{bmatrix} = \begin{bmatrix} 2Z & -Z \\ -Z & 2Z \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$\Delta = 3\Delta^2$$

$$\Delta_1 = (208)(1.5 + j0.866)Z$$

$$\Delta_2 = (208)(j1.732)Z$$

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{(208)(1.5 + j0.866)}{(3)(60 - j30)} = 1.789\angle 56.56^\circ$$

$$I_2 = \frac{\Delta_2}{\Delta} = \frac{(208)(j1.732)}{(3)(60 - j30)} = 1.79\angle 116.56^\circ$$

$$P_1 = R_e[V_1 I_1^*] = R_e[(208)(1.789\angle -56.56^\circ)]$$

$$= 208.98 \text{ W}$$

$$P_2 = R_e[V_2 I_2^*] = R_e[(208)\angle -60^\circ \times 1.79\angle 63.44^\circ]$$

$$= 371.65 \text{ W}$$